Gradient Boosting Regression

* Additive Modelling

$$F(x) = f_0(x) + f_1(x) + f_2(x) \dots f_n(x)$$

* Algorithm

Input -

1) Training set $\{(x_i, y_i)\}_{i=1}^{n}$

2) A differentiable loss function

$L(y, F(x)) \Rightarrow L(y, \hat{y})$,

3) Number of iterations M

Steps

1) Initialize $f_0(x) = \arg\min_{\gamma} \sum_{i=1}^{N} L(y_i, \gamma)$

Our original Loss Function is

$$L = \frac{1}{2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Now in terms of γ

$$f_0(x) = \arg\min_{\gamma} \boxed{\frac{1}{2} \sum_{i=1}^{n} (y_i - \gamma)^2}$$

This can be read as, we need a that value of γ, for which value of expression (highlighted by yellow) should be min

diff w.r.t γ

$$d\, f_0(x) = d \quad 1 \quad \sum^{n} (y - \gamma)^2$$

$$\frac{\quad}{d\gamma} \qquad \frac{\quad}{d\gamma} \; \frac{\quad}{2} \; i=1$$

$$= \frac{1}{2} \sum_{i=1}^{n} \frac{d}{d\gamma} (y_i - \gamma)^2$$

$$= \frac{1}{\cancel{2}} \times \cancel{2} (y_i - \gamma) \times \frac{d}{d\gamma} (y_i - \gamma)$$

$$= (y_i - \gamma) \times (-1)$$

$$= -\sum_{i=1}^{n} (y_i - \gamma) = 0$$

$$= \sum_{i=1}^{n} (\gamma - y_i) = 0$$

Considering one dataset

R&D Spend	Administration	Marketing spend	Profit
165	137	472	192

101	92	250	144
29	127	201	91

$$\sum_{i=1}^{3} (\gamma - y_i) = 0$$

$$\Rightarrow (\gamma - 192) + (\gamma - 144) + (\gamma - 91) = 0$$

$$\Rightarrow 3\gamma - 192 - 144 - 91 = 0$$

$$\Rightarrow \gamma = \frac{192 + 144 + 91}{3}$$

Which is equal to mean of output column

2) For $m = 1$ to M:

This is loop

$$F(x) = f_0(x) + f_1(x) + f_2(x) + \dots + f_n(x)$$

$f_0(x) \Rightarrow$ Already calculated

For remaining terms i.e. $f_1(x) \ldots F_n(x)$ loop is used.

(a) For $i = 1, 2, \ldots, N$ compute

$$r_{im} = -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f = f_{m-1}}$$

r = residual / Pseudo residual

i = row

m = Decision Tree

$$r_{i1} = -\left[\frac{\partial}{\partial \hat{y}_i} L(y_i, \hat{y})\right]$$

$$L = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$r_{i1} = -\left[\frac{\partial}{\partial \hat{y}_i} \frac{1}{2} (y_i - \hat{y}_i)^2\right]_{f=f_0}$$

$$= \left[(y_i - \hat{y}_i)\right]_{f=f_0}$$

$$= \left[(y_i - f(x_i)\right]_{f=f_0} \quad \because \hat{y}_i = f(x_i)$$

$$r_{i1} = (y_i - f_0(x_i))$$

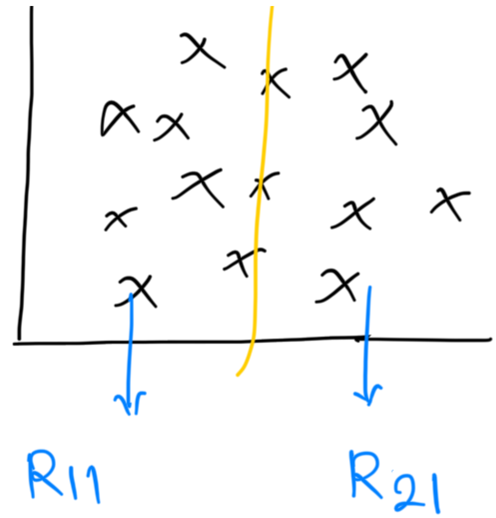
$$r_{11} = y_1 - f_0(x_1) = 192 - 142 = 50$$

$$r_{21} = y_2 - f(x_2) = 144 - 142 = 2$$

$$r_{31} = y_3 - f(x_3) = 91 - 142 = -51$$

(b) Fit a regression tree to the targets r_{im} giving terminal regions R_{jm}, $j = 1, 2, \ldots, J_m$

Terminal Region -



R_{11} = 1^{st} DT's 1^{st} terminal region

R_{21} = 1^{st} DT's 2^{nd} terminal region

(c) For $j = 1, 2 \ldots J_m$ compute

$$\gamma_{jm} = \arg\min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, f_{m-1}(x_i) + \gamma)$$

$$\gamma_{11} = \arg\min_{\gamma} \frac{1}{2}(y_i - F_0(x_i) + \gamma)^2$$

$$\frac{dL}{\partial \gamma} = \frac{1}{2} \times 2(y_i - f_0(x) + \gamma)\frac{d}{d\gamma}(y_i - f_0(x) - \gamma) = 0$$

$$= (y_i - f_0(x) - \gamma) = 0$$

$$\Rightarrow y_i - f_0(x) - \gamma) = 0$$

$$\gamma_{11} = 91 - 142 - \gamma = 0$$

$$\therefore \gamma = 91 - 142 = -51$$

$$\gamma_{21} = \underset{\gamma}{\arg\min} \sum_{x_i \in R_{21}} L(y_i, f_0(x_i) + \gamma)$$

$$= \underset{\gamma}{\arg\min} \frac{1}{2} \sum_{i=1}^{2} (y_i - f_0(x_i) + \gamma)^2$$

$$= -\sum_{i=1}^{2} (y_i - f_0(x_i) - \gamma) = 0$$

$$= \sum_{i=1}^{2} (y_i - f_0(x_i) - \gamma = 0$$

$$= y_1 - f_0(x_1) - \gamma + \gamma \quad - f_0(x_2) - \gamma = 0$$

$$\Rightarrow 192 - 142 - \gamma + 144 - 142 - \gamma = 0$$

$$\therefore \gamma = 26$$

(d) Update $f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} I(x \in R_{jm})$

$f_1(x) = f_0(x) + dT_1$

$f_2(x) = f_1(x) + dT_2$

$\vdots$

$f_n(x) = f_{n-1}(x) + dT_n$

3) Output $\hat{f}(x) = f_M(x)$